E-commerce & Customer Behavior Analytics

Project Documentation

Project Category	Business Intelligence and Data Analytics
Tool	Microsoft Power BI
Dataset	Olist Brazilian E-Commerce Public Dataset
Data Format	CSV (Comma-Separated Values)
Reporting Deliverable	Three-page Power BI report and one separate Power BI Service dashboard

1. Project Overview

E-commerce & Customer Behavior Analytics is a business intelligence project developed using Microsoft Power BI to analyze online retail transactions, customer purchasing behavior, order trends, and product category performance.

The project uses the Olist Brazilian E-Commerce Public Dataset, which contains information about customers, orders, order items, products, and product categories. The data is transformed into a structured analytical model and visualized through interactive reports and an executive dashboard.

The solution converts raw transactional data into meaningful business insights using Power Query, data modeling, DAX measures, and data visualization.

The final solution consists of three report pages:

- E-commerce Sales Overview
- Customer Behavior & Segmentation
- Product Performance

A separate Power BI Service dashboard, named E-commerce Executive Dashboard, provides a consolidated view of important business indicators.

2. Problem Statement

E-commerce businesses generate large amounts of transactional data every day. However, raw data stored in multiple CSV files and database tables does not automatically provide meaningful insights into business performance.

Without an integrated analytics solution, businesses may face difficulties in:

- Monitoring revenue and order volume.
- Identifying changes in monthly sales performance.
- Understanding customer distribution across geographic regions.
- Measuring repeat-purchase behavior.
- Identifying high-performing product categories.
- Comparing product demand with revenue contribution.
- Presenting important business metrics to management in a centralized view.

The project addresses these challenges by transforming multiple datasets into an integrated Power BI reporting solution that supports data-driven decision-making.

3. Objectives of the Project

The primary objective is to develop an interactive business intelligence solution for e-commerce sales and customer analytics.

The specific objectives are:

- Import and prepare e-commerce data from CSV files.
- Clean and validate the data using Power Query.
- Establish relationships between customers, orders, products, and order items.
- Create DAX measures for revenue, orders, customers, average order value, repeat customers, and items sold.
- Analyze monthly revenue trends.

- Compare customer distribution across states.
- Measure repeat-customer and one-time-customer behavior.
- Identify product categories that generate high revenue or sell high volumes.
- Build three analytical report pages.
- Publish the report to Power BI Service.
- Create a separate executive dashboard using pinned report visuals.
- Develop practical skills in business intelligence, data modeling, and analytical reporting.

4. Business Problem Solved

The project provides a centralized view of important e-commerce performance indicators.

Business challenge	Project solution	Business benefit
Difficult to monitor overall performance	KPI cards for revenue, orders, customers, and average order value	Faster performance assessment
Limited visibility into sales trends	Monthly revenue line chart	Helps identify changes over time
Difficult to compare customer locations	Customers-by-state bar chart	Supports geographic market analysis
Limited understanding of customer retention	Repeat-customer and one-time-customer measures	Helps evaluate repeat-purchase behavior
Difficult to identify important product categories	Revenue bar chart and category treemap	Supports product portfolio decisions
Revenue and product volume can tell different stories	Comparison of revenue and items sold	Helps distinguish high-value and high-volume categories
Business metrics are spread across report pages	Separate executive dashboard	Provides a consolidated business overview

The solution supports decision-making; it does not automatically prescribe actions or prove the causes of changes in business performance.

5. Business Perspective

From a business perspective, the project focuses on four major areas.

5.1 Sales Performance

Sales performance analysis helps managers understand revenue generation, order volume, and average order value.

Business questions include:

- What is the total product sales revenue?
- How does revenue change month by month?
- How many orders were recorded?
- What is the average product sales revenue per order?

5.2 Customer Analytics

Customer analytics focuses on understanding customer reach, geographic distribution, and repeat purchasing.

Business questions include:

- How many unique customer identities are represented in the dataset?
- Which states have the largest customer counts?
- How many customers have placed multiple orders?
- What proportion of customers are repeat customers?

5.3 Product Performance

Product performance analysis compares category-level sales value and item volume.

Business questions include:

- Which product categories generate the most revenue?
- Which categories have the highest item volumes?
- Which categories contribute most to overall product sales?
- Do high-volume categories also generate high revenue?

5.4 Executive Reporting

The dashboard presents selected KPIs and charts in a single view so managers can quickly assess business performance before opening the detailed report for further exploration.

6. Scope of the Project

Included in scope

- CSV data import.
- Data preparation and data type validation.
- Duplicate-order diagnostic checks.
- Relational data modeling.
- DAX measure development.
- Sales trend analysis.
- Customer distribution and repeat-purchase analysis.
- Product category revenue and item-volume analysis.
- Three-page Power BI report.
- Publishing to Power BI Service.
- Separate executive dashboard.

Outside the current scope

- Real-time transaction processing.
- Predictive sales forecasting.
- Customer lifetime value modeling.
- Profitability analysis using complete cost data.
- Automated marketing recommendations.
- Live integration with an e-commerce platform.
- Advanced machine learning.

These features can be considered future enhancements.

7. Dataset Description

The project uses the Olist Brazilian E-Commerce Public Dataset, commonly distributed through Kaggle.

The dataset contains multiple CSV files representing different aspects of e-commerce operations. The project uses five principal tables.

Dataset/table	Important fields	Purpose
orders	order_id, customer_id, order_status, order_purchase_timestamp	Order tracking and time-based analysis
order_items	order_id, product_id, price, freight_value	Product sales value and item-volume calculations
customers	customer_id, customer_unique_id, customer_city, customer_state, customer_zip_code_prefix	Customer identification and geographic analysis
products	product_id, product_category_name	Product and category analysis
product_category_name_translation	product_category_name, product_category_name_english	English category labels

The dataset also contains other files, such as payments, reviews, sellers, and geolocation information. These were not required for the core implementation.

7.1 Dataset format

The source data is provided in CSV format.

CSV is a plain-text tabular format in which each row represents a record and columns represent attributes.

Example:

order_id	customer_id	order_status
ORD001	CUST001	delivered
ORD002	CUST002	delivered

The values above are illustrative examples, not original Olist records.

7.2 Data types

Appropriate data types were assigned during data preparation:

- IDs: Text.
- Timestamps: Date/Time.
- Prices and freight values: Decimal numbers.
- Customer ZIP-code prefix: Text.
- Order status and state: Text.

Identifiers should generally be treated as identifiers rather than quantities. For example, adding ZIP-code prefixes has no analytical meaning.

8. Tools and Technologies Used

Tool or technology	Role in the project
Microsoft Power BI Desktop	Report development and data modeling
Power Query	Data cleaning and transformation
DAX	Measures and analytical calculations
Power BI data model	Relationships between tables
Power BI visualizations	Charts, KPI cards, and category comparisons
Power BI Service	Publishing and dashboard creation
CSV files	Source data format

9. Project Methodology

The project follows a structured business intelligence workflow.

Phase 1: Data Acquisition

The Olist CSV files were downloaded and imported into Power BI Desktop.

The required tables were selected based on the analytical objectives.

Phase 2: Data Cleaning and Transformation

Power Query was used to inspect the imported data, promote column headers where necessary, validate data types, and prepare the data for modeling.

A separate diagnostic query, Check_duplicate_orders, was created to group records by order_id and count their occurrences.

The duplicate check was used to identify possible data-quality issues. It was not intended to remove legitimate order-item records.

Phase 3: Data Modeling

Relationships were established between the main tables using their common identifiers.

The model connects customers to orders, orders to order items, products to order items, and products to their English category translations.

The relationships enable Power BI to apply filters across related tables.

Phase 4: DAX Development

DAX measures were created to calculate the main KPIs and customer behavior indicators.

The measures were then reused across report visuals rather than writing separate calculations for every chart.

Phase 5: Report Development

The report was organized into three pages, each with a distinct analytical purpose:

- Sales overview.
- Customer behavior.
- Product performance.

Phase 6: Publishing and Dashboard Development

The report was saved as a Power BI Desktop .pbix file and published to Power BI Service.

Selected visuals were pinned to a separate dashboard named E-commerce Executive Dashboard.

Phase 7: Validation

The report and dashboard should be checked for correct measure results, working relationships, readable labels, and consistent definitions of revenue and customers.

10. Data Model and Relationships

The model uses relationships between the following columns:

- customers[customer_id] → orders[customer_id]
- orders[order_id] → order_items[order_id]
- products[product_id] → order_items[product_id]
- product_category_name_translation[product_category_name] → products[product_category_name]

The intended cardinalities are one-to-many from customers to orders, orders to order items, products to order items, and category translations to products.

The category translation table should have one row per Portuguese category name for the last relationship to remain one-to-many. Duplicate category keys must be investigated rather than forcing an incorrect relationship.

11. DAX Measures and Their Purpose

11.1 Total Revenue

```
Total Revenue =
SUM(order_items[price])
```

Calculates the sum of product prices across order-item rows.

Business use: Sales KPI, revenue trend analysis, and category revenue comparisons.

Limitation: The measure excludes freight, costs, and profit calculations. Unless the business rules are refined, it can include items associated with cancelled or unavailable orders.

11.2 Total Orders

```
Total Orders =
DISTINCTCOUNT(orders[order_id])
```

Counts unique order IDs.

Business use: Measures order volume without counting multiple items in the same order as separate orders.

11.3 Total Customers

```
Total Customers =
DISTINCTCOUNT(customers[customer_unique_id])
```

Counts distinct customer identities.

Business use: Customer reach and customer-count reporting.

11.4 Average Order Value

```
Average Order Value =
DIVIDE(
    [Total Revenue],
    [Total Orders],
    0
)
```

Divides total product sales revenue by the number of unique orders.

Business use: Evaluates average product sales value per order.

11.5 Repeat Customers

```
Repeat Customers =
COUNTROWS(
    FILTER(
        VALUES(customers[customer_unique_id]),
        CALCULATE(
            DISTINCTCOUNT(orders[order_id])
        ) > 1
    )
)
```

Counts customer identities associated with more than one distinct order under the current data model and filters.

Business use: Repeat-purchase and customer-retention analysis.

11.6 One-Time Customers

```
One-Time Customers =
COUNTROWS(
    FILTER(
        VALUES(customers[customer_unique_id]),
        CALCULATE(
            DISTINCTCOUNT(orders[order_id])
        ) = 1
    )
)
```

Counts customer identities associated with exactly one distinct order.

Business use: Helps compare first-time or single-order customers with repeat customers.

11.7 Repeat Customer Rate

```
Repeat Customer Rate =
DIVIDE(
    [Repeat Customers],
    [Total Customers],
    0
)
```

Calculates repeat customers divided by total customers.

Business use: Expresses repeat purchasing as a proportion of the selected customer population.

The measure should be formatted as a percentage. For rigorous analysis, the numerator and denominator should use compatible customer populations and clearly defined order-status rules.

11.8 Total Items Sold

```
Total Items Sold =
COUNTROWS(order_items)
```

Counts the rows in the order-items table.

Business use: Measures item-row volume and enables comparisons between sales value and quantity.

It is an item-row count, not a distinct product count. The Olist order-items data does not require a separate quantity multiplication for this calculation because each row represents an order item.

11.9 Customer Loyalty Table

```
Customer Loyalty =
DATATABLE(
    "Customer Type", STRING,
    {
        {"Repeat Customers"},
        {"One-Time Customers"}
    }
)
```

Creates a small table containing two customer categories for the donut chart.

11.10 Customer Loyalty Count

```
Customer Loyalty Count =
SWITCH(
    SELECTEDVALUE('Customer Loyalty'[Customer Type]),
    "Repeat Customers", [Repeat Customers],
    "One-Time Customers", [One-Time Customers],
    0
)
```

Returns the appropriate customer measure based on the selected customer type.

The customer loyalty table is intentionally disconnected. The measure maps each category to the corresponding calculation.

12. Report Page Design

Page 1: E-commerce Sales Overview

Purpose: Provide a high-level view of business performance.

Visual	Type	Main field or measure	Reason for selection
Total Revenue	Card	[Total Revenue]	Highlights product sales value
Average Order Value	Card	[Average Order Value]	Shows average product sales value per order
Total Customers	Card	[Total Customers]	Summarizes customer reach
Total Orders	Card	[Total Orders]	Summarizes order volume
Monthly Revenue Trend	Line chart	Purchase month and [Total Revenue]	Reveals revenue changes over time

Page 2: Customer Behavior & Segmentation

Purpose: Analyze geographic distribution and customer loyalty.

Visual	Type	Main field or measure	Reason for selection
Customers by State	Horizontal bar chart	State and distinct customer count	Compares geographic customer distribution
Customer Loyalty	Donut chart	Customer type and [Customer Loyalty Count]	Shows the share of repeat and one-time customers
Total Customers	Card	[Total Customers]	Provides overall customer count
Repeat Customers	Card	[Repeat Customers]	Highlights repeat purchasing
One-Time Customers	Card	[One-Time Customers]	Highlights single-order customers
Repeat Customer Rate	Card	[Repeat Customer Rate]	Summarizes repeat-customer proportion

Page 3: Product Performance

Purpose: Compare category-level item volume and revenue.

Visual	Type	Main field or measure	Reason for selection
Total Items Sold	Card	[Total Items Sold]	Summarizes item-row volume
Items Sold by Category	Clustered column chart	English category and [Total Items Sold]	Compares item volume by category
Category Revenue Share	Treemap	English category and [Total Revenue]	Shows relative category contribution
Revenue by Product Category	Horizontal bar chart	English category and [Total Revenue]	Ranks categories by revenue

The English category field is supplied by product_category_name_translation.

13. Power BI Service Dashboard

The separate dashboard is named E-commerce Executive Dashboard.

It was created in Power BI Service by pinning selected report visuals.

Recommended dashboard content includes:

- Total Revenue.
- Total Orders.
- Total Customers.
- Average Order Value.
- Monthly Revenue Trend.
- Repeat Customer Rate.
- Revenue by Product Category.

The dashboard provides a single-page summary of selected indicators. The underlying report remains the place for detailed analysis and full report filtering.

A dashboard tile may be a snapshot or have behavior that differs from the original report visual. Dashboard interactions should therefore be validated in Power BI Service.

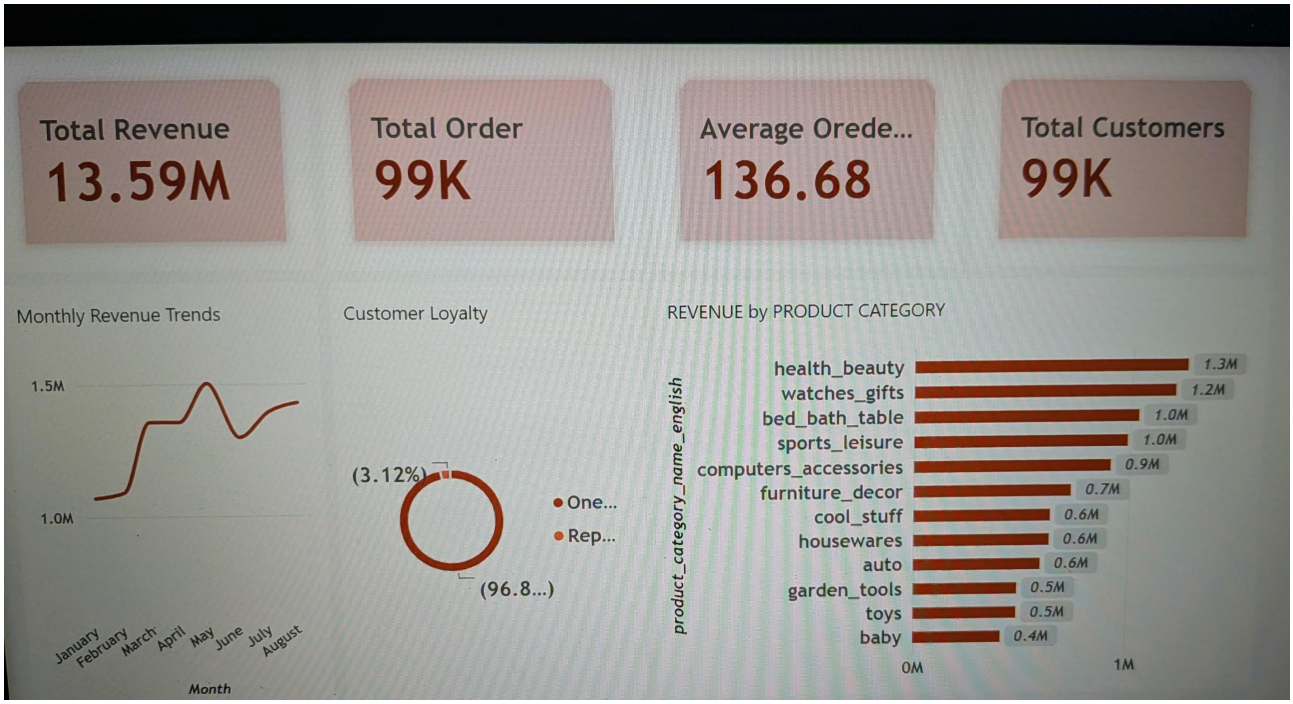


Figure 1: E-commerce Executive Dashboard – Total Revenue, Total Orders, Average Order Value, Total Customers, Monthly Revenue Trends, Customer Loyalty and Revenue by Product Category.

14. Key Findings and Business Interpretation

The report is designed to help users investigate the following areas:

Sales trends

The monthly revenue line chart helps identify changes in sales value over time. Further investigation is required to determine the causes of increases or decreases.

Customer distribution

The customers-by-state chart highlights geographic concentration and may help identify regions for further market analysis.

Customer loyalty

Repeat-customer measures help estimate the extent of repeat purchasing. A low rate may justify further investigation, but does not by itself explain why customers do not return.

Product category performance

The revenue bar chart and treemap help identify categories contributing more to product sales value.

Sales volume versus sales value

The item-volume chart allows categories with high item counts to be compared against categories with high revenue.

The project is descriptive analytics. It summarizes historical data and supports investigation rather than forecasting future outcomes or establishing causation.

15. Applications and Real-World Use Cases

This solution can be adapted for:

- E-commerce businesses: Track revenue, order volume, product categories, and repeat purchasing.

- Retail companies: Compare product demand and sales performance across categories.
- Marketing teams: Identify customer segments and geographic markets for further investigation.
- Business analysts: Prepare recurring KPI reports and performance summaries.
- Operations teams: Examine order trends and investigate unusual changes.
- Management teams: Use the executive dashboard to review business indicators.
- Academic projects: Demonstrate data preparation, relational modeling, DAX, and dashboard development.

The dataset represents historical Brazilian e-commerce activity. Applying the solution to another business would require adapting the data model, metrics, and definitions to that business's data.

16. Learning Objectives

The project provides practical experience in the following areas:

- Understanding CSV-based business datasets.
- Importing and transforming data using Power Query.
- Working with identifiers, timestamps, and numeric data types.
- Detecting potential data-quality issues.
- Designing one-to-many relationships.
- Understanding facts, dimensions, and relational modeling.
- Writing DAX measures.
- Using aggregation and filtering functions.
- Designing KPI cards, line charts, bar charts, column charts, donut charts, and treemaps.
- Choosing visualizations based on analytical questions.
- Publishing reports to Power BI Service.
- Creating dashboards by pinning report visuals.
- Interpreting business metrics and documenting limitations.
- Presenting analytical findings in a business-friendly format.

17. Project Limitations

The current solution has several limitations:

- Revenue definition: Total Revenue sums item prices and excludes freight, operating costs, and profit calculations.
- Order-status treatment: The core revenue measure does not automatically exclude cancelled or unavailable orders.
- Repeat-customer denominator: The repeat-customer rate should be validated to ensure numerator and denominator use compatible customer populations.
- Item quantity: Total Items Sold counts order-item rows rather than distinct products or separately recorded quantities.